

# Sequences and Basic Series Convergence Tests

ANSWER KEY



## Convergence of sequences

- 1 Determine whether the sequence  $a_n = \frac{3n+1}{n}$  converges, and if so, find its limit.

$a_n = 3 + \frac{1}{n} \rightarrow 3$  as  $n \rightarrow \infty$ . Converges to 3.

- 2 Determine whether the sequence  $a_n = (-1)^n$  converges.

The sequence oscillates between  $-1$  and  $1$  and never settles on a single value, so it diverges.

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## Geometric series

- 3 Determine whether  $\sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^n$  converges, and if so, find its sum.

Geometric series with  $r = \frac{1}{2}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{3}{1-1/2} = 6$ .

- 4 Determine whether  $\sum_{n=1}^{\infty} 2^n$  converges.

Geometric with  $r = 2$ ,  $|r| \geq 1$ : diverges.

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## The $n$ th-term (divergence) test

- 5 Use the  $n$ th-term test to show  $\sum_{n=1}^{\infty} \frac{n}{n+1}$  diverges.

$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0$ . Since the terms don't approach 0, the series diverges by the  $n$ th-term test.

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## The integral test

- 6 Use the Integral Test to determine whether  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  converges.

Let  $f(x) = \frac{1}{x^2}$ , positive, continuous, decreasing for  $x \geq 1$ .  $\int_1^{\infty} \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_1^{\infty} = 0 - (-1) = 1$ , a finite value.  
By the Integral Test, the series converges.

- 7 Use the Integral Test to determine whether  $\sum_{n=1}^{\infty} \frac{1}{n}$  converges (the harmonic series).

$\int_1^{\infty} \frac{1}{x} dx = [\ln x]_1^{\infty} = \infty$ . Diverges. So the harmonic series diverges by the Integral Test.

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## The comparison and limit comparison tests

- 8 Use the Direct Comparison Test to determine whether  $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$  converges, comparing to  $\sum \frac{1}{n^2}$ .  
 $\frac{1}{n^2+1} < \frac{1}{n^2}$  for all  $n \geq 1$ , and  $\sum \frac{1}{n^2}$  converges (p-series with  $p = 2 > 1$ ). By Direct Comparison,  $\sum \frac{1}{n^2+1}$  converges.
- 9 Use the Limit Comparison Test to determine whether  $\sum_{n=1}^{\infty} \frac{2n}{n^2+3}$  converges, comparing to  $\sum \frac{1}{n}$ .  
 $\lim_{n \rightarrow \infty} \frac{\frac{2n}{n^2+3}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{2n^2}{n^2+3} = 2$ , a finite positive value. Since  $\sum \frac{1}{n}$  diverges (harmonic),  $\sum \frac{2n}{n^2+3}$  also diverges by the Limit Comparison Test.

## Visualizing sequence convergence

- 10 Plot the first 6 terms of  $a_n = \frac{3n+1}{n}$  (from the earlier question) and describe how the points behave as  $n$  increases.

$a_1 = 4$ ,  $a_2 = 3.5$ ,  $a_3 \approx 3.33$ ,  $a_4 = 3.25$ ,  $a_5 = 3.2$ ,  $a_6 \approx 3.17$ . The points decrease and cluster tighter and tighter around the horizontal line  $y = 3$ , visually confirming the limit found algebraically.

## Monotonic and bounded sequences

- 11 Determine whether  $a_n = \frac{n}{n+1}$  is monotonic (always increasing or always decreasing), and find an upper bound.

$a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0$ , so the sequence is strictly increasing. Since  $a_n = 1 - \frac{1}{n+1} < 1$  for all  $n$ , it is bounded above by 1.

- 12 State the Monotonic Sequence Theorem, and use it to conclude that  $a_n = \frac{n}{n+1}$  converges.

The Monotonic Sequence Theorem states that every bounded, monotonic sequence converges. Since  $a_n = \frac{n}{n+1}$  is increasing and bounded above by 1 (shown above), it must converge (in fact to 1).

## The p-series test

- 13 State the p-series test for  $\sum_{n=1}^{\infty} \frac{1}{n^p}$ , and classify convergence for  $p = 1$ ,  $p = \frac{3}{2}$ , and  $p = \frac{1}{2}$ .

$\sum \frac{1}{n^p}$  converges if  $p > 1$  and diverges if  $p \leq 1$ .  $p = 1$ : diverges (harmonic).  $p = \frac{3}{2} > 1$ : converges.  $p = \frac{1}{2} \leq 1$ : diverges.

- 14 Use the p-series test (after simplifying) to determine whether  $\sum_{n=1}^{\infty} \frac{5}{n^{0.9}}$  converges.

This is  $5 \sum \frac{1}{n^{0.9}}$ , a constant multiple of a p-series with  $p = 0.9 \leq 1$ . Diverges.

## Series review: choosing the right test

15 For each series, name the most efficient test to apply (do not fully evaluate):

a  $\sum \frac{1}{5^n}$  Geometric series test ( $r = \frac{1}{5}$ )

b  $\sum \frac{1}{n^3}$  p-series test ( $p = 3$ )

c  $\sum \frac{n+1}{2n+3}$  nth-term test (terms  $\rightarrow \frac{1}{2} \neq 0$ )

d  $\sum \frac{1}{n^2 + 2n}$  Direct/Limit Comparison to  $\sum \frac{1}{n^2}$

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## Sequences defined recursively

16 A sequence is defined recursively by  $a_1 = 2$ ,  $a_{n+1} = \frac{a_n + 6}{2}$ . Compute  $a_2$ ,  $a_3$ , and  $a_4$ , and conjecture the limit if the sequence converges.

$a_2 = \frac{2+6}{2} = 4$ .  $a_3 = \frac{4+6}{2} = 5$ .  $a_4 = \frac{5+6}{2} = 5.5$ . The terms appear to approach 6 (if  $L = \frac{L+6}{2}$ , then  $2L = L + 6 \Rightarrow L = 6$ ).

17 Verify that if a recursively defined sequence  $a_{n+1} = \frac{a_n + 6}{2}$  converges to a limit  $L$ , then  $L$  must satisfy  $L = 6$ .

If  $a_n \rightarrow L$ , then  $a_{n+1} \rightarrow L$  as well, so taking the limit of both sides:  $L = \frac{L+6}{2} \Rightarrow 2L = L + 6 \Rightarrow L = 6$ , confirming the conjecture.

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## Geometric series with algebraic manipulation

18 Find the sum of  $\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n$  (note the series starts at  $n = 2$ , not  $n = 0$ ).

Using  $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$  and subtracting the first two terms ( $n = 0, 1$ ):  $\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n = \frac{1}{1-\frac{2}{3}} - 1 - \frac{2}{3} = 3 - \frac{5}{3} = \frac{4}{3}$ .

19 Express the repeating decimal  $0.\overline{45}$  as a geometric series and find its exact fractional value.

$0.454545\dots = \sum_{n=1}^{\infty} 45 \left(\frac{1}{100}\right)^n = \frac{45/100}{1-1/100} = \frac{45/100}{99/100} = \frac{45}{99} = \frac{5}{11}$ .

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## Telescoping series

20 Evaluate the telescoping series  $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$  by examining its partial sums.

$S_N = \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{N+1}$  (all middle terms cancel). As  $N \rightarrow \infty$ :  $S_N \rightarrow 1 - 0 = 1$ . The series converges to 1.

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## Multiple choice — easy 15

21 Find the sum of  $\sum_{n=0}^{\infty} 9\left(-\frac{1}{3}\right)^n$ .

a. 12

b.  $\frac{31}{4}$

c.  $\frac{27}{4}$  ✓

d.  $\frac{27}{2}$

Geometric series with  $r = -\frac{1}{3}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{9}{1-(-\frac{1}{3})} = \frac{27}{4}$ .

22 Find the sum of  $\sum_{n=0}^{\infty} 3\left(\frac{1}{6}\right)^n$ .

a.  $\frac{23}{5}$

b.  $\frac{5}{2}$

c.  $\frac{18}{7}$

d.  $\frac{18}{5}$  ✓

Geometric series with  $r = \frac{1}{6}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{3}{1-(\frac{1}{6})} = \frac{18}{5}$ .

23 Find the sum of  $\sum_{n=0}^{\infty} 2\left(\frac{2}{3}\right)^n$ .

a. 7

b.  $\frac{6}{5}$

c. 6 ✓

d.  $\frac{2}{3}$

Geometric series with  $r = \frac{2}{3}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{2}{1-(\frac{2}{3})} = 6$ .

24 Find the sum of  $\sum_{n=0}^{\infty} 5\left(\frac{1}{6}\right)^n$ .

a. 6 ✓

b.  $\frac{30}{7}$

c. 7

d.  $\frac{25}{6}$

Geometric series with  $r = \frac{1}{6}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{5}{1-(\frac{1}{6})} = 6$ .

25 Find the sum of  $\sum_{n=0}^{\infty} 6\left(\frac{1}{2}\right)^n$ .

a. 12 ✓

b. 13

c. 4

d. 3

Geometric series with  $r = \frac{1}{2}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{6}{1-\left(\frac{1}{2}\right)} = 12$ .

26 Find the sum of  $\sum_{n=0}^{\infty} 9\left(\frac{2}{5}\right)^n$ .

a. 16

b. 15 ✓

c.  $\frac{45}{7}$

d.  $\frac{27}{5}$

Geometric series with  $r = \frac{2}{5}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{9}{1-\left(\frac{2}{5}\right)} = 15$ .

27 Find the sum of  $\sum_{n=0}^{\infty} 3\left(\frac{1}{4}\right)^n$ .

a. 4 ✓

b.  $\frac{9}{4}$

c. 5

d.  $\frac{12}{5}$

Geometric series with  $r = \frac{1}{4}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{3}{1-\left(\frac{1}{4}\right)} = 4$ .

28 Find the sum of  $\sum_{n=0}^{\infty} 7\left(\frac{1}{2}\right)^n$ .

- a. 15
- b.  $\frac{7}{2}$
- c.  $\frac{14}{3}$
- d. 14 ✓

Geometric series with  $r = \frac{1}{2}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{7}{1-\left(\frac{1}{2}\right)} = 14$ .

29 Find the sum of  $\sum_{n=0}^{\infty} 3\left(-\frac{1}{2}\right)^n$ .

- a. 3
- b. 6
- c.  $\frac{9}{2}$
- d. 2 ✓

Geometric series with  $r = -\frac{1}{2}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{3}{1-\left(-\frac{1}{2}\right)} = 2$ .

30 Find the sum of  $\sum_{n=0}^{\infty} 1\left(\frac{1}{3}\right)^n$ .

- a.  $\frac{3}{2}$  ✓
- b.  $\frac{5}{2}$
- c.  $\frac{3}{4}$
- d.  $\frac{2}{3}$

Geometric series with  $r = \frac{1}{3}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{1}{1-\left(\frac{1}{3}\right)} = \frac{3}{2}$ .

31 Find the sum of  $\sum_{n=0}^{\infty} 8\left(\frac{1}{6}\right)^n$ .

- a.  $\frac{48}{7}$
- b.  $\frac{48}{5}$  ✓
- c.  $\frac{53}{5}$
- d.  $\frac{20}{3}$

Geometric series with  $r = \frac{1}{6}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{8}{1-\left(\frac{1}{6}\right)} = \frac{48}{5}$ .

32 Find the sum of  $\sum_{n=0}^{\infty} 1\left(\frac{2}{3}\right)^n$ .

a. 4

b. 3 ✓

c.  $\frac{1}{3}$

d.  $\frac{3}{5}$

Geometric series with  $r = \frac{2}{3}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{1}{1-\left(\frac{2}{3}\right)} = 3$ .

33 Find the sum of  $\sum_{n=0}^{\infty} 4\left(\frac{2}{5}\right)^n$ .

a.  $\frac{20}{7}$

b.  $\frac{23}{3}$

c.  $\frac{12}{5}$

d.  $\frac{20}{3}$  ✓

Geometric series with  $r = \frac{2}{5}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{4}{1-\left(\frac{2}{5}\right)} = \frac{20}{3}$ .

34 Find the sum of  $\sum_{n=0}^{\infty} 1\left(\frac{2}{5}\right)^n$ .

a.  $\frac{5}{7}$

b.  $\frac{3}{5}$

c.  $\frac{8}{3}$

d.  $\frac{5}{3}$  ✓

Geometric series with  $r = \frac{2}{5}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{1}{1-\left(\frac{2}{5}\right)} = \frac{5}{3}$ .

35 Find the sum of  $\sum_{n=0}^{\infty} 7\left(\frac{1}{6}\right)^n$ .

a.  $\frac{35}{6}$

b.  $\frac{42}{5}$  ✓

c. 6

d.  $\frac{47}{5}$

Geometric series with  $r = \frac{1}{6}$ ,  $|r| < 1$ : converges. Sum  $= \frac{a}{1-r} = \frac{7}{1-\left(\frac{1}{6}\right)} = \frac{42}{5}$ .

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### Multiple choice — medium 15

36 Find  $\lim_{n \rightarrow \infty} \frac{-6n+9}{7n+3}$ .

a.  $\frac{1}{7}$

b.  $\frac{6}{7}$

c.  $-\frac{7}{6}$

d.  $-\frac{6}{7}$  ✓

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{-6+9/n}{7+3/n} = \frac{-6}{7} = -\frac{6}{7}$ .

37 Find  $\lim_{n \rightarrow \infty} \frac{9n+9}{-8n+4}$ .

a.  $-\frac{9}{8}$  ✓

b.  $\frac{9}{8}$

c.  $-\frac{1}{8}$

d.  $-\frac{8}{9}$

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{9+9/n}{-8+4/n} = \frac{9}{-8} = -\frac{9}{8}$ .

38 Find  $\lim_{n \rightarrow \infty} \frac{6n-6}{-7n+2}$ .

a.  $-\frac{7}{6}$

b.  $\frac{6}{7}$

c.  $\frac{1}{7}$

d.  $-\frac{6}{7}$  ✓

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{6-6/n}{-7+2/n} = \frac{6}{-7} = -\frac{6}{7}$ .

39 Find  $\lim_{n \rightarrow \infty} \frac{-6n-5}{-2n-7}$ .

a.  $-3$

b.  $4$

c.  $3$  ✓

d.  $\frac{1}{3}$

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{-6-5/n}{-2-7/n} = \frac{-6}{-2} = 3$ .

40 Find  $\lim_{n \rightarrow \infty} \frac{-n-5}{6n-8}$ .

a.  $-\frac{1}{6}$  ✓

b.  $-6$

c.  $\frac{5}{6}$

d.  $\frac{1}{6}$

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{-1-5/n}{6-8/n} = \frac{-1}{6} = -\frac{1}{6}$ .

41 Find  $\lim_{n \rightarrow \infty} \frac{8n-7}{-8n+7}$ .

a.  $-\frac{1}{2}$

b.  $1$

c.  $0$

d.  $-1$  ✓

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{8-7/n}{-8+7/n} = \frac{8}{-8} = -1$ .

42 Find  $\lim_{n \rightarrow \infty} \frac{3n+2}{-6n-3}$ .

a.  $-2$

b.  $-1$

c.  $\frac{1}{2}$

d.  $-\frac{1}{2}$  ✓

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{3+2/n}{-6-3/n} = \frac{3}{-6} = -\frac{1}{2}$ .

43 Find  $\lim_{n \rightarrow \infty} \frac{5n-3}{9n+1}$ .

a.  $-\frac{5}{9}$

b.  $\frac{14}{9}$

c.  $\frac{9}{5}$

d.  $\frac{5}{9}$  ✓

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{5-3/n}{9+1/n} = \frac{5}{9} = \frac{5}{9}$ .

44 Find  $\lim_{n \rightarrow \infty} \frac{-5n+4}{6n-2}$ .

a.  $-\frac{6}{5}$

b.  $\frac{5}{6}$

c.  $-\frac{5}{6}$  ✓

d.  $\frac{1}{6}$

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{-5+4/n}{6-2/n} = \frac{-5}{6} = -\frac{5}{6}$ .

45 Find  $\lim_{n \rightarrow \infty} \frac{-7n-9}{8n+3}$ .

- a.  $\frac{1}{8}$
- b.  $\frac{7}{8}$
- c.  $-\frac{8}{7}$
- d.  $-\frac{7}{8}$  ✓

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{-7-9/n}{8+3/n} = \frac{-7}{8} = -\frac{7}{8}$ .

46 Find  $\lim_{n \rightarrow \infty} \frac{2n+9}{8n+9}$ .

- a.  $-\frac{1}{4}$
- b. 4
- c.  $\frac{1}{4}$  ✓
- d.  $\frac{5}{4}$

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{2+9/n}{8+9/n} = \frac{2}{8} = \frac{1}{4}$ .

47 Find  $\lim_{n \rightarrow \infty} \frac{6n+7}{-7n-4}$ .

- a.  $-\frac{7}{6}$
- b.  $\frac{6}{7}$
- c.  $\frac{1}{7}$
- d.  $-\frac{6}{7}$  ✓

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{6+7/n}{-7-4/n} = \frac{6}{-7} = -\frac{6}{7}$ .

48 Find  $\lim_{n \rightarrow \infty} \frac{5n-2}{-6n-3}$ .

- a.  $\frac{1}{6}$
- b.  $-\frac{6}{5}$
- c.  $-\frac{5}{6}$  ✓
- d.  $\frac{5}{6}$

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{5-2/n}{-6-3/n} = \frac{5}{-6} = -\frac{5}{6}$ .

49 Find  $\lim_{n \rightarrow \infty} \frac{8n-6}{-6n+8}$ .

a.  $-\frac{4}{3}$  ✓

b.  $-\frac{1}{3}$

c.  $-\frac{3}{4}$

d.  $\frac{4}{3}$

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{8-6/n}{-6+8/n} = \frac{8}{-6} = -\frac{4}{3}$ .

50 Find  $\lim_{n \rightarrow \infty} \frac{-2n+5}{n+9}$ .

a.  $-2$  ✓

b.  $2$

c.  $-1$

d.  $-\frac{1}{2}$

Dividing numerator and denominator by  $n$ :  $\lim_{n \rightarrow \infty} \frac{-2+5/n}{1+9/n} = \frac{-2}{1} = -2$ .

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### Multiple choice — hard 15

51 Find the sum of  $\sum_{n=2}^{\infty} 5\left(\frac{2}{5}\right)^n$  (note the series starts at  $n = 2$ ).

a.  $\frac{10}{3}$

b.  $\frac{25}{3}$

c.  $\frac{4}{3}$  ✓

d.  $\frac{7}{3}$

$$\sum_{n=2}^{\infty} 5\left(\frac{2}{5}\right)^n = 5\left(\frac{2}{5}\right)^2 \cdot \frac{1}{1-\left(\frac{2}{5}\right)} = \frac{4}{3}.$$

52 Find the sum of  $\sum_{n=1}^{\infty} 4\left(-\frac{2}{3}\right)^n$  (note the series starts at  $n = 1$ ).

a.  $\frac{12}{5}$

b.  $-\frac{21}{10}$

c.  $-\frac{3}{5}$

d.  $-\frac{8}{5}$  ✓

$$\sum_{n=1}^{\infty} 4\left(-\frac{2}{3}\right)^n = 4\left(-\frac{2}{3}\right)^1 \cdot \frac{1}{1 - \left(-\frac{2}{3}\right)} = -\frac{8}{5}.$$

53 Find the sum of  $\sum_{n=1}^{\infty} 8\left(-\frac{1}{3}\right)^n$  (note the series starts at  $n = 1$ ).

a.  $-2$  ✓

b. 6

c.  $-1$

d.  $-\frac{5}{2}$

$$\sum_{n=1}^{\infty} 8\left(-\frac{1}{3}\right)^n = 8\left(-\frac{1}{3}\right)^1 \cdot \frac{1}{1 - \left(-\frac{1}{3}\right)} = -2.$$

54 Find the sum of  $\sum_{n=2}^{\infty} 2\left(\frac{2}{3}\right)^n$  (note the series starts at  $n = 2$ ).

a.  $\frac{8}{3}$  ✓

b. 4

c. 6

d.  $\frac{11}{3}$

$$\sum_{n=2}^{\infty} 2\left(\frac{2}{3}\right)^n = 2\left(\frac{2}{3}\right)^2 \cdot \frac{1}{1 - \left(\frac{2}{3}\right)} = \frac{8}{3}.$$

55 Find the sum of  $\sum_{n=2}^{\infty} 9\left(\frac{2}{5}\right)^n$  (note the series starts at  $n = 2$ ).

a. 15

b.  $\frac{17}{5}$

c.  $\frac{12}{5}$  ✓

d. 6

$$\sum_{n=2}^{\infty} 9\left(\frac{2}{5}\right)^n = 9\left(\frac{2}{5}\right)^2 \cdot \frac{1}{1 - \left(\frac{2}{5}\right)} = \frac{12}{5}.$$

56 Use the  $p$ -series test to classify  $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{2}{3}}}$ .

a. Converges ( $p = \frac{2}{3} > 1$ )

b. **Diverges** ( $p = \frac{2}{3} \leq 1$ ) ✓

c. Diverges (terms don't approach 0, by the  $n$ th-term test)

d. Converges (geometric series test)

This is a  $p$ -series with  $p = \frac{2}{3}$ . Since  $p \leq 1$ : Diverges ( $p = \frac{2}{3} \leq 1$ ).

57 Use the  $p$ -series test to classify  $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$ .

a. Converges ( $p = \frac{1}{2} > 1$ )

b. Converges (geometric series test)

c. **Diverges** ( $p = \frac{1}{2} \leq 1$ ) ✓

d. Diverges (terms don't approach 0, by the  $n$ th-term test)

This is a  $p$ -series with  $p = \frac{1}{2}$ . Since  $p \leq 1$ : Diverges ( $p = \frac{1}{2} \leq 1$ ).

58 Use the  $p$ -series test to classify  $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ .

a. Converges (geometric series test)

b. **Converges** ( $p = \frac{3}{2} > 1$ ) ✓

c. Diverges ( $p = \frac{3}{2} \leq 1$ )

d. Diverges (terms don't approach 0, by the  $n$ th-term test)

This is a  $p$ -series with  $p = \frac{3}{2}$ . Since  $p > 1$ : Converges ( $p = \frac{3}{2} > 1$ ).

59 Use the  $p$ -series test to classify  $\sum_{n=1}^{\infty} \frac{1}{n^2}$ .

- a. Diverges ( $p = 2 \leq 1$ )
- b. **Converges ( $p = 2 > 1$ ) ✓**
- c. Converges (geometric series test)
- d. Diverges (terms don't approach 0, by the  $n$ th-term test)

This is a  $p$ -series with  $p = 2$ . Since  $p > 1$ : Converges ( $p = 2 > 1$ ).

60 Use the  $p$ -series test to classify  $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{4}{3}}}$ .

- a. **Converges ( $p = \frac{4}{3} > 1$ ) ✓**
- b. Diverges (terms don't approach 0, by the  $n$ th-term test)
- c. Diverges ( $p = \frac{4}{3} \leq 1$ )
- d. Converges (geometric series test)

This is a  $p$ -series with  $p = \frac{4}{3}$ . Since  $p > 1$ : Converges ( $p = \frac{4}{3} > 1$ ).

61 Use the Limit Comparison Test on  $\sum_{n=1}^{\infty} \frac{4n}{6n^2 - 9}$ , comparing to  $\sum \frac{1}{n}$ . Find  $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ .

- a.  $-\frac{2}{3}$
- b.  $\frac{3}{2}$
- c.  $\frac{2}{3}$  ✓
- d.  $\frac{5}{3}$

$\frac{a_n}{b_n} = \frac{4n}{6n^2 - 9} \cdot n = \frac{4n^2}{6n^2 - 9}$ . As  $n \rightarrow \infty$ , this ratio  $\rightarrow \frac{4}{6} = \frac{2}{3}$ .

62 Use the Limit Comparison Test on  $\sum_{n=1}^{\infty} \frac{6n}{5n^2}$ , comparing to  $\sum \frac{1}{n}$ . Find  $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ .

- a.  $\frac{6}{5}$  ✓
- b.  $\frac{5}{6}$
- c.  $\frac{11}{5}$
- d.  $-\frac{6}{5}$

$\frac{a_n}{b_n} = \frac{6n}{5n^2} \cdot n = \frac{6n^2}{5n^2}$ . As  $n \rightarrow \infty$ , this ratio  $\rightarrow \frac{6}{5} = \frac{6}{5}$ .

63 Use the Limit Comparison Test on  $\sum_{n=1}^{\infty} \frac{5n}{7n^2+4}$ , comparing to  $\sum \frac{1}{n}$ . Find  $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ .

a.  $-\frac{5}{7}$

b.  $\frac{7}{5}$

c.  $\frac{12}{7}$

d.  $\frac{5}{7}$  ✓

$$\frac{a_n}{b_n} = \frac{5n}{7n^2+4} \cdot n = \frac{5n^2}{7n^2+4}. \text{ As } n \rightarrow \infty, \text{ this ratio } \rightarrow \frac{5}{7} = \frac{5}{7}.$$

64 Use the Limit Comparison Test on  $\sum_{n=1}^{\infty} \frac{7n}{3n^2+9}$ , comparing to  $\sum \frac{1}{n}$ . Find  $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ .

a.  $\frac{7}{3}$  ✓

b.  $-\frac{7}{3}$

c.  $\frac{3}{7}$

d.  $\frac{10}{3}$

$$\frac{a_n}{b_n} = \frac{7n}{3n^2+9} \cdot n = \frac{7n^2}{3n^2+9}. \text{ As } n \rightarrow \infty, \text{ this ratio } \rightarrow \frac{7}{3} = \frac{7}{3}.$$

65 Use the Limit Comparison Test on  $\sum_{n=1}^{\infty} \frac{n}{5n^2+6}$ , comparing to  $\sum \frac{1}{n}$ . Find  $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ .

a.  $\frac{6}{5}$

b.  $\frac{1}{5}$  ✓

c. 5

d.  $-\frac{1}{5}$

$$\frac{a_n}{b_n} = \frac{n}{5n^2+6} \cdot n = \frac{n^2}{5n^2+6}. \text{ As } n \rightarrow \infty, \text{ this ratio } \rightarrow \frac{1}{5} = \frac{1}{5}.$$